

PART VIII – BIG-O ANALYSIS

Students must analyze and explain, in their own words, why the following complexities apply to their implementation. This assessment does not require complexity analysis beyond what has been covered in Weeks 1–5. Write your explanations in the accompanying Student Documentation Worksheet.

Operation	Time Complexity	Explain Why (student to complete)
Direct array access	$O(1)$	The program accesses a student directly using an array index, so only one position needs to be accessed.
Array traversal	$O(n)$	displayStudents() uses a loop to go through all student records from the first student to the last.
Linear search	$O(n)$	Searching, updating, and deleting a student checks the student IDs one by one until the matching ID is found.
Matrix traversal ($n \times n$)	$O(n^2)$	processMatrix() uses two nested loops. One loop goes through the rows and the other goes through the columns, so every matrix element is checked.